

João Pedro Paiva Pires



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I studied Physics, ended up spending most of my time building ML models, and now I'm doing an MSc in Data Science at IST. Along the way I co-authored a paper on battery monitoring with TCN models, currently under review at Nature Machine Intelligence, and applied deep learning models to classify PSI20 movements that won Best Project in my course. I'm looking for a summer internship in quant research or data science at a bank or asset manager.

Education

Instituto Superior Técnico, MSc in Data Science and Engineering Sept 2025 – July 2027

- **Coursework:** Machine Learning, Deep Learning, Computational Statistics, Network Science, Information Systems and Databases, Information Processing and Retrieval, Time Series Analysis.

University of Aveiro, BSc in Physics Sept 2022 – July 2025

- **Coursework:** Calculus, Linear Algebra, Discrete Mathematics, Numerical Methods, Classical Mechanics, Statistical Physics, Quantum Mechanics, Electrodynamics, Solid State Physics, Atomic and Nuclear Physics.
- **Research Grant Award** – Recognized with a research grant during my final year, declined due to institutional exclusivity.

Publications

"Beyond the Surface: Estimating Internal Battery Pressure and Temperature with Optical Fiber Sensors and Machine Learning", Nature Machine Intelligence *Under-Review*

L. Matuck, M. Nascimento, J. P. Pires, J. F. F. Mendes, M. S. Ferreira

- Developed preprocessing pipelines and tuned TCN architectures via Bayesian optimization (500 trials) to reconstruct internal battery states from non-invasive optical fiber signals.
- Achieved R^2 of 0.996 (temperature) and 0.97 (pressure) for LIB, contributing as an undergraduate to doctoral-level research submitted to Nature Machine Intelligence.

Projects

Deep Learning Architectures for Predicting PSI20 Trends github.com/joaopires

- Framed index movement prediction as a 3-class problem (up/flat/down) and benchmarked LSTM, Attention-LSTM, and Transformer architectures on 10 years of daily price and volume data, achieving Macro F1 of 62.5%.
- Built a preprocessing pipeline combining Wavelet Transform denoising with Z-score normalization to reduce noise in financial time series prior to model training.
- Optimized architecture hyperparameters via Bayesian Search. Awarded **Best Project (18/20)**, self-initiated outside of curriculum.

Personal Finance Tracker github.com/joaopires

- Built an end-to-end pipeline in R and PostgreSQL: automated transaction ingestion, iterative ML classifier (Naive Bayes → Logistic Regression) for category labeling, and a 3-model forecast ensemble (Holt-Winters, Linear Trend, Moving Average) for monthly spending prediction.
- Deployed a live Shiny dashboard with real-time CRUD operations and NLP-based category suggestions.

Extracurricular Activities

Mentor in the UA Mentorship Program, University of Aveiro – Aveiro, PT Sept 2024 – June 2025

- Mentored 15 first-year students throughout the full academic year, guiding them on course selection, study strategies, and career path orientation through regular sessions.
- Developed leadership and communication skills by fostering a supportive environment that helped students navigate both academic and personal challenges of university life.

Technical Skills & Interests

Programming Languages & Databases: Python, R, SQL (PostgreSQL), Matlab.

ML Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Hugging Face.

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly.

Interests: Systematic trading strategies, factor investing, financial time series, quantitative finance.